SMART INDIA HACKATHON 2026

APAAR-VERIFY

- **Problem Statement ID – SIH26263**
- **Problem Statement Title- APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- [Add Team ID]**
- **Team Name- Team Phoenix**



Team Phoenix

APAAR-VERIFY

SMART INDIA HACKATHON 2025

Proposed Solution

- **Unified APAAR platform:** institutions submit student-level records keyed by APAAR ID, not aggregate totals
- **Automated validation:** catches invalid IDs, duplicates, excess intake and multi-institution enrolment
- **Anomaly detection:** flags enrolment spikes and NRI/OCI/FN patterns; risk score per institution
- **Real-time dashboards:** role-based intake vs. enrolment, alerts and reports
- **Tamper-proof verification:** digitally signed QR records; employer checks need student consent
- **Security by design:** JWT, RBAC, encryption, audit trails, DPDP Act 2023 aligned

How it addresses the problem

- Replaces unverifiable totals with auditable records
- Stops ghost and duplicate enrolments early
- Gives AICTE live visibility of seat caps

Innovation & uniqueness

- Only system combining APAAR-keyed data, cross-institution duplicate checks and risk scoring
- Consent-based verification with signed QR

Team Phoenix

TECHNICAL APPROACH

SMART INDIA HACKATHON 2025

Methodology: end-to-end workflow

1 Upload
Institution submits CSV / API data

2 Validate
ID, duplicate, intake checks with row-level errors

3 Store
Verified records saved to central repository

4 Monitor
Dashboards, alerts, risk scores for AICTE

5 Verify
Signed QR check with student consent

Technologies to be used

Frontend

- React.js, React Router
- Tailwind CSS, Bootstrap
- Recharts, Chart.js

Backend

- Node.js, Express.js (REST)
- Multer for CSV upload
- Socket.IO for live alerts

Database & storage

- MongoDB, Mongoose
- MongoDB Atlas
- Cloudinary

Security & cloud

- JWT, bcrypt, RBAC, Nodemailer
- Docker, GitHub, Postman
- Vercel, Render, AWS

Live prototype: https://apaar-verify-sih.vercel.app/login | Integrations: APAAR API, AICTE Portal, DigiLocker/NAD

Team Phoenix

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2025

Feasibility

- **Technical:** proven React, Node, MongoDB, JWT stack
- **Modular:** phased rollout, upload to verification
- **Policy:** APAAR ID backed by NEP 2020
- **Economic:** scalable, low-cost cloud

Viability

- **Scale:** thousands of institutions, millions of students
- **Reuse:** extends to UGC and State Councils
- **Revenue:** per-verification fee, analytics, ERP licensing

Challenges and mitigation

Area	Challenge	Strategy
Privacy	Student data exposure	Encryption, RBAC, audit logs, consent (DPDP Act 2023)
Adoption	Institutions resist regular uploads	Standard CSV templates, row-level error feedback
Interoperability	Varied campus ERP systems	REST APIs with a pluggable APAAR adapter
Data quality	Incomplete or duplicate data	Pre-commit validation and anomaly detection

Team Phoenix

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2025

Impact on target audience

AICTE: Real-time intake vs. enrolment, seat-cap alerts, ghost enrolment detection

Institutions: Bulk upload with instant error feedback

Employers: Instant, tamper-evident credential checks with consent

Students: One portfolio, control over who can view it

Benefits of the solution

Social	Authentic credentials, less fraud
Economic	Lower verification cost, no scholarship leakage
Administrative	Reliable accreditation data, easy tracking
Technological	Faster APAAR adoption via secure APIs
Environmental	Paperless, digital verification

Team Phoenix

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

Comparison with existing systems

Feature	APAAR-Verify	AICTE Portal	AISHE	DigiLocker
Student-level data by APAAR ID	**Yes**	No	No	Partial
Intake vs. enrolment check	**Yes**	Partial	No	No
Cross-institution duplicates	**Yes**	No	No	No
Real-time dashboards	**Yes**	Partial	No	No
Anomaly detection, risk score	**Yes**	No	No	No
Employer verification	**Yes**	No	No	Yes
Signed QR records	**Yes**	No	No	Partial

References

- **APAAR ID:** 12-digit lifelong student ID under NEP 2020, launched 13 Feb 2024
- **APAAR portal:** apaar.education.gov.in
- **AICTE:** aicte-india.org
- **DigiLocker:** digilocker.gov.in
- **Ministry of Education:** education.gov.in
- **Brochure:** NCERT CIET APAAR Brochure (PDF)